cool off to space. Water vapor, though, absorbs at infrared frequencies that correspond in part to the windows in the carbon dioxide opacity. The two gases together, it seemed to me, could pretty well absorb almost all the infrared emission, even if there was very little water vapor—something like two picket fences, the slats of one being fortuitously positioned to cover the gaps of the other.

There was another very different category of explanation, in which the high brightness temperature of Venus had nothing to do with the ground. The surface could still be temperate, clement, congenial. It was proposed that some region in the atmosphere of Venus or in its surrounding magnetosphere emitted these radio waves to space. Electrical discharges between water droplets in the Venus clouds were suggested. A glow discharge in which ions and electrons recombined at twilight and dawn in the upper atmosphere was offered. A very dense ionosphere had its advocates, in which the mutual acceleration of unbound electrons ("free-free emission") gave off radio waves. (One proponent of this idea even suggested that the high ionization required was due to an average of 10,000 times greater radioactivity on Venus than on Earth—perhaps from a recent nuclear war there.) And, in the light of the discovery of radiation from Jupiter's magnetosphere, it was natural to suggest that the radio emission came from an immense cloud of charged particles trapped by some hypothetical very intense Venusian magnetic field.

In a series of papers I published in the middle 1960s, many in collaboration with Jim Pollack,* these conflicting models of a high hot emitting region and a cold surface were subjected to a critical analysis. By then we had two important new clues: the radio spectrum of Venus, and the *Mariner 2* evidence that the radio emission was more intense at the center of the disk of Venus than toward its edge. By 1967 we were able to exclude the alternative models with some confidence, and conclude that the surface of Venus was at a scorching and un-Earthlike temperature, in excess of 400°C. But the argument was inferential, and there were many intermediate steps. We longed for a more direct measurement.